

IP ADDRESSING AND SUBNETTING

Task – 1 IP address allocating.

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Class	Default Subnet Mask	First Octet Range	Real-World Example & Use Case
Class A	255.0.0.0	1 – 126	Large Enterprise or ISP Infrastructure: Very large networks like Apple or IBM (e.g., 17.0.0.0/8 assigned to Apple) or private enterprise backbones using the 10.0.0.0/8 private range for thousands of corporate devices.
Class B	255.255.0.0	128 – 191	University Campus Network or Hospital System: Medium-to-large networks, such as a university dorm network or hospital IT system using the 172.16.0.0/12 private range or assigned public blocks.
Class C	255.255.255.0	192 – 223	Home Wi-Fi Router / Personal Devices: Small local networks (LANs), such as a home router assigning local IP addresses (e.g., 192.168.1.100) to laptops, smart TVs, and smartphones.
Class D		224 – 239	IPTV Streams & Routing Protocols: Video streaming services broadcasting live cable channels to multiple subscribers simultaneously (e.g., 239.255.0.1), or routing updates via OSPF/RIP.

Class E		240 – 255	Network R&D Laboratories: Reserved exclusively for experimental purposes, Internet Engineering Task Force (IETF) research, and internal TCP/IP stack testing.
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Task – 2 Network address and the boardcast.

- 1. Binary Conversion & Bitmask Identification
IP Address (192.168.10.45): 11000000.10101000.00001010.00101101
- Subnet Mask (255.255.255.0): 11111111.11111111.11111111.00000000 (/24 CIDR notation)
- The mask reserves the first 24 bits (3 octets) for the network portion and the remaining 8 bits (1 octet) for host addresses.
- 2. Network Address Calculation :- Perform a bitwise AND operation between the IP address and the subnet mask:
- Octet 1: 192 AND 255 = 192
- Octet 2: 168 AND 255 = 168
- Octet 3: 10 AND 255 = 10
- Octet 4: 45 AND 0 = 0 (00101101 AND 00000000 = 00000000)
- Network Address: 192.168.10.0
- 3. Broadcast Address Calculation :- Keep the network bits intact and set all host bits (the last octet) to binary 1s (11111111 = 255 decimal):
- Octet 1: 192
- Octet 2: 168
- Octet 3: 10
- Octet 4: 255
- Broadcast Address: 192.168.10.255
- 4. Valid Host Address Range Calculation :- The usable host range falls strictly between the network address and the broadcast address:
- First Valid Host: Network Address + 1 = 192.168.10.1
- Last Valid Host: Broadcast Address – 1 = 192.168.10.254
- Valid Host Range: 192.168.10.1 – 192.168.10.254 (254 usable host IPs total)

Task – 3 Allocating the ip address to different works.

Subnet Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Task – 4 Calculating the different submask.

- 1. Determine the Number of Borrowed Bits :- To create at least 20 subnets, calculate the required borrowed subnet bits (n) using the formula $2^n \geq 20$:
- $2^4 = 16$ subnets (*insufficient*)
- $2^5 = 32$ subnets (*satisfies the requirement of ≥ 20*)
- 2. Calculate the New Subnet Mask
- Original Prefix: /16 (255.255.0.0)
- New Prefix Length: $16 + 5 = /21$
- Binary Mask: 11111111.11111111.11110000.00000000
- Decimal Conversion: 255.255.248.0
- 3. Calculate the Number of Hosts per Subnet
- Remaining Host Bits (h): $32 - 21 = 11$ bits
- Total Addresses per Subnet: $2^{11} = 2,048$
- Usable Host Addresses: $2^{11} - 2 = 2,046$ (*subtracting network and broadcast addresses*)
- Final Answers :- New Subnet Mask: 255.255.248.0 (or /21)
- Subnets Created: 32 subnets

- Usable Hosts per Subnet: 2,046 hosts

Task – 5 Problem of connecting through the IP address.

- The error is that 192.168.1.300 is an invalid IPv4 address. In IPv4 networking, an IP address consists of four numbers (octets) separated by periods, and each octet must fall within the range of 0 to 255. The last octet in your friend's address is 300, which exceeds the maximum limit of an 8-bit binary number ($2^8 - 1 = 255$). Assuming a standard home network setup (192.168.1.0/24 with subnet mask 255.255.255.0):
 - Valid Host Range: 192.168.1.1 to 192.168.1.254
 - Example Valid IP Address: 192.168.1.30 (or any available number between 1 and 254 for the last octet).
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